

7.27 (a) With $\vec{B} = \nabla \times \vec{A}$, we can write the integrand as

$$\begin{aligned}\vec{x} \times (\vec{E} \times \vec{B}) &= \vec{x} \times [\vec{E} \times (\nabla \times \vec{A})] = \epsilon_{abc} \hat{e}_a x_b [\vec{E} \times (\nabla \times \vec{A})]_c \\ &= \epsilon_{abc} \hat{e}_a x_b \epsilon_{cde} E_d (\nabla \times \vec{A})_e = \epsilon_{abc} \hat{e}_a x_b \epsilon_{cde} E_d \epsilon_{efg} \partial_f A_g \\ &= \epsilon_{abc} (\delta_{ef} \delta_{dg} - \delta_{eg} \delta_{df}) \hat{e}_a x_b E_d \partial_f A_g \\ &= E_d (\epsilon_{abc} \hat{e}_a x_b \partial_c) A_d - \epsilon_{abc} \hat{e}_a x_b (E_d \partial_d) A_c\end{aligned}$$

The first term is $\sum_{j=1}^3 E_j (\vec{x} \times \nabla) A_j$, and the second term can be expressed as

$$\epsilon_{abc} \hat{e}_a (E_d \partial_d (x_b A_c) - E_d A_c \delta_{bd}) = (\vec{E} \cdot \nabla) (\vec{x} \times \vec{A}) - \vec{E} \times \vec{A}.$$

$$\text{Therefore, } \vec{x} \times (\vec{E} \times \vec{B}) = \vec{E} \times \vec{A} + \sum_{j=1}^3 E_j (\vec{x} \times \nabla) A_j - (\vec{E} \cdot \nabla) (\vec{x} \times \vec{A}).$$

Since for an arbitrary function $f(\vec{x})$, we have

$$\begin{aligned}\int (\vec{E} \cdot \nabla) f(\vec{x}) d^3x &= \sum_{i=1}^3 \int E_i (\partial_i f) d^3x = \sum_{i=1}^3 \int d^3x_{\perp,i} \int dx_i E_i (\partial_i f) \\ &= \sum_{i=1}^3 \int d^3x_{\perp,i} \left(E_i f(x_i, x_{\perp,i}) \Big|_{x_i=x_{\min}}^{x_{\max}} \right) - \sum_{i=1}^3 \int d^3x (\partial_i E_i) f \\ &= \oint (\vec{E} \cdot \vec{n}) f da - \int (\nabla \cdot \vec{E}) f d^3x.\end{aligned}$$

$$\text{So } \int (\vec{E} \cdot \nabla) (\vec{x} \times \vec{A}) d^3x = \oint (\vec{E} \cdot \vec{n}) (\vec{x} \times \vec{A}) da - \int (\nabla \cdot \vec{E}) (\vec{x} \times \vec{A}) d^3x$$

If the field and potential fall fast enough, the surface integral can be dropped. Also, $\nabla \cdot \vec{E} = 0$,

when there is no source. Finally, we will have

$$\vec{L} = \frac{1}{\mu_0 c} \int [\vec{x} \times (\vec{E} \times \vec{B})] d^3x = \frac{1}{\mu_0 c} \int [\vec{E} \times \vec{A} + \sum_{j=1}^3 E_j (\vec{x} \times \nabla) A_j] d^3x$$

(b) In the radiation gauge,

$$\vec{E} = -\frac{\partial \vec{A}}{\partial t} = \sum_{\lambda} \int \frac{d^3k}{(2\pi)^3} i\omega \left[\vec{E}_{\lambda}(\vec{k}) a_{\lambda}(\vec{k}) e^{i\vec{k}\cdot\vec{x} - i\omega t} - \vec{E}_{\lambda}^*(\vec{k}) a_{\lambda}^*(\vec{k}) e^{-i\vec{k}\cdot\vec{x} + i\omega t} \right]$$

Since $\vec{E}_+ \times \vec{E}_+ = \vec{E}_- \times \vec{E}_- = \vec{E}_+ \times \vec{E}_-^* = \vec{E}_- \times \vec{E}_+^* = 0$, $\vec{E}_+ \times \vec{E}_- = \vec{E}_- \times \vec{E}_+^* = -i\vec{E}_3$,

$\vec{E}_- \times \vec{E}_+ = \vec{E}_- \times \vec{E}_-^* = i\vec{E}_3$, the spin part of the angular momentum becomes

$$\begin{aligned} L_{spin} &= \frac{1}{\mu_0 c^2} \int d^3x \sum_{\lambda} \sum_{\lambda'} \int \frac{d^3k}{(2\pi)^3} \int \frac{d^3k'}{(2\pi)^3} i\omega \left[\vec{E}_{\lambda}(\vec{k}) a_{\lambda}(\vec{k}) e^{i\vec{k}\cdot\vec{x} - i\omega t} - \vec{E}_{\lambda}^*(\vec{k}) a_{\lambda}^*(\vec{k}) e^{-i\vec{k}\cdot\vec{x} + i\omega t} \right] \\ &\quad \times \left[\vec{E}_{\lambda'}(\vec{k}') a_{\lambda'}(\vec{k}') e^{i\vec{k}'\cdot\vec{x} - i\omega' t} + \vec{E}_{\lambda'}^*(\vec{k}') a_{\lambda'}^*(\vec{k}') e^{-i\vec{k}'\cdot\vec{x} + i\omega' t} \right] \\ &= \frac{1}{\mu_0 c^2} \int d^3x \int \frac{d^3k}{(2\pi)^3} \int \frac{d^3k'}{(2\pi)^3} i\omega \left[-i\vec{E}_3 a_+(\vec{k}) a_-(\vec{k}') e^{i(\vec{k}+\vec{k}')\cdot\vec{x} - i(\omega+\omega')t} - i\vec{E}_3 a_+(\vec{k}) a_+^*(\vec{k}') e^{i(\vec{k}-\vec{k}')\cdot\vec{x} - i(\omega-\omega')t} \right. \\ &\quad + i\vec{E}_3 a_-(\vec{k}) a_+(\vec{k}') e^{i(\vec{k}+\vec{k}')\cdot\vec{x} - i(\omega+\omega')t} + i\vec{E}_3 a_-(\vec{k}) a_+^*(\vec{k}') e^{i(\vec{k}-\vec{k}')\cdot\vec{x} - i(\omega-\omega')t} \\ &\quad - i\vec{E}_3 a_+^*(\vec{k}) a_+(\vec{k}') e^{-i(\vec{k}-\vec{k}')\cdot\vec{x} + i(\omega-\omega')t} - i\vec{E}_3 a_+^*(\vec{k}) a_+^*(\vec{k}') e^{-i(\vec{k}+\vec{k}')\cdot\vec{x} + i(\omega+\omega')t} \\ &\quad \left. + i\vec{E}_3 a_+^*(\vec{k}) a_-(\vec{k}') e^{-i(\vec{k}-\vec{k}')\cdot\vec{x} + i(\omega-\omega')t} + i\vec{E}_3 a_+^*(\vec{k}) a_+^*(\vec{k}') e^{-i(\vec{k}+\vec{k}')\cdot\vec{x} + i(\omega+\omega')t} \right] \\ &= \frac{1}{\mu_0 c^2} \int \frac{d^3k}{(2\pi)^3} i\omega \left[-i\vec{E}_3 a_+(\vec{k}) a_-(\vec{k}) e^{-2i\omega t} - i\vec{E}_3 a_+(\vec{k}) a_+^*(\vec{k}) + i\vec{E}_3 a_-(\vec{k}) a_+(\vec{k}) e^{-2i\omega t} + i\vec{E}_3 a_-(\vec{k}) a_+^*(\vec{k}) \right. \\ &\quad \left. - i\vec{E}_3 a_+^*(\vec{k}) a_+(\vec{k}) - i\vec{E}_3 a_+^*(\vec{k}) a_+^*(\vec{k}) e^{2i\omega t} + i\vec{E}_3 a_+^*(\vec{k}) a_-(\vec{k}) + i\vec{E}_3 a_+^*(\vec{k}) a_+^*(\vec{k}) e^{2i\omega t} \right] \\ &= \frac{2}{\mu_0 c} \int \frac{d^3k}{(2\pi)^3} \vec{k} \left[|a_+(\vec{k})|^2 + a_-(\vec{k}) a_+(-\vec{k}) e^{-2i\omega t} - |a_-(\vec{k})|^2 - a_+^*(\vec{k}) a_+^*(-\vec{k}) e^{2i\omega t} \right] \end{aligned}$$

Taking the time average, the oscillating terms proportional to $e^{-2i\omega t}$ and $e^{2i\omega t}$ will become zero.

Therefore
$$L_{spin} = \frac{2}{\mu_0 c} \int \frac{d^3k}{(2\pi)^3} \vec{k} \left[|a_+(\vec{k})|^2 - |a_-(\vec{k})|^2 \right]$$

The energy can be calculated from $U = \frac{\epsilon_0}{2} \int d^3x |\vec{E}|^2$. Notice that

$$\vec{E}_+ \cdot \vec{E}_+ = \vec{E}_- \cdot \vec{E}_- = \vec{E}_+ \cdot \vec{E}_-^* = \vec{E}_- \cdot \vec{E}_+^* = 0, \quad \vec{E}_+ \cdot \vec{E}_- = \vec{E}_+ \cdot \vec{E}_+^* = \vec{E}_- \cdot \vec{E}_-^* = 1,$$

We have

$$\begin{aligned} U &= \frac{\epsilon_0}{2} \int d^3x \sum_{\lambda} \sum_{\lambda'} \int \frac{d^3k}{(2\pi)^3} \int \frac{d^3k'}{(2\pi)^3} (-\omega\omega') \left[\vec{E}_{\lambda}(k) a_{\lambda}(k) e^{i\vec{k} \cdot \vec{x} - i\omega t} - \vec{E}_{\lambda}^*(k) a_{\lambda}^*(k) e^{-i\vec{k} \cdot \vec{x} + i\omega t} \right] \\ &\quad \cdot \left[\vec{E}_{\lambda'}(k') a_{\lambda'}(k') e^{i\vec{k}' \cdot \vec{x} - i\omega' t} - \vec{E}_{\lambda'}^*(k') a_{\lambda'}^*(k') e^{-i\vec{k}' \cdot \vec{x} + i\omega' t} \right] \\ &= \frac{\epsilon_0}{2} \int d^3x \int \frac{d^3k}{(2\pi)^3} \int \frac{d^3k'}{(2\pi)^3} (-\omega\omega') \left[a_+(k) a_-(k') e^{i(\vec{k}+\vec{k}') \cdot \vec{x} - i(\omega+\omega')t} - a_+(k) a_+^*(k') e^{i(\vec{k}-\vec{k}') \cdot \vec{x} - i(\omega-\omega')t} \right. \\ &\quad + a_-(k) a_+(k') e^{i(\vec{k}+\vec{k}') \cdot \vec{x} - i(\omega+\omega')t} - a_-(k) a_-^*(k') e^{i(\vec{k}-\vec{k}') \cdot \vec{x} - i(\omega-\omega')t} \\ &\quad - a_+^*(k) a_+(k') e^{-i(\vec{k}-\vec{k}') \cdot \vec{x} + i(\omega-\omega')t} + a_+^*(k) a_-^*(k') e^{-i(\vec{k}+\vec{k}') \cdot \vec{x} + i(\omega+\omega')t} \\ &\quad \left. - a_-^*(k) a_-(k') e^{-i(\vec{k}-\vec{k}') \cdot \vec{x} + i(\omega-\omega')t} + a_-^*(k) a_+^*(k') e^{-i(\vec{k}+\vec{k}') \cdot \vec{x} + i(\omega+\omega')t} \right] \\ &= \frac{\epsilon_0}{2} \int \frac{d^3k}{(2\pi)^3} \omega^2 \left[-a_+(k) a_-(-k) e^{-2i\omega t} + a_+(k) a_+^*(k) - a_-(k) a_+(-k) e^{-2i\omega t} + a_-(k) a_-^*(k) \right. \\ &\quad \left. + a_+^*(k) a_+(k) - a_+^*(k) a_-^*(-k) e^{2i\omega t} + a_-^*(k) a_-(k) - a_-^*(k) a_+^*(-k) e^{2i\omega t} \right] \\ &= \epsilon_0 \int \frac{d^3k}{(2\pi)^3} \omega^2 \left[|a_+(k)|^2 - a_+(k) a_-(-k) e^{-2i\omega t} + |a_-(k)|^2 - a_+^*(k) a_-^*(-k) e^{2i\omega t} \right] \end{aligned}$$

Again, time average leads to

$$U = \epsilon_0 \int \frac{d^3k}{(2\pi)^3} \omega^2 \left[|a_+(k)|^2 + |a_-(k)|^2 \right]$$